

Velum.

Holder-attested compliance for Confidential Tokens on Stellar.

A regulated fund can now hide its register of holders *and* prove it obeys the rules — identity-gated confidential tokens, plus on-chain proof that a position clears a regulatory threshold, without revealing the amount to anyone. Including the verifier.

Compliance has two halves. Upstream ships the primitives for both, wired to neither.

THE QUESTION	UPSTREAM TODAY	VELUM
Who may hold?	A flat address allowlist . No reason, no expiry, no attesting party — insufficient for a securities regime. Bridging to the RWA identity registry is their issue #766, still open .	<code>velum-policy</code> — the token asks an ERC-3643/T-REX identity registry : does this holder carry a valid qualified-investor claim from an approved issuer?
How much may they hold?	Structurally impossible . <code>is_authorized(account, token) -> bool</code> sees no value — balances are Pedersen commitments. The RWA modules that enforce caps demand cleartext <code>amount: i128</code> . The two cannot meet.	<code>disclose_balance_ge</code> + <code>velum-attest</code> — the holder proves “my $position \geq T$ ” and a Soroban contract verifies it. Nothing is decrypted; there is nothing to decrypt.

Why the check had to move

Under FHE a contract compares ciphertexts with no witness, so the check fits inside the transfer. Under Pedersen commitments a proof *needs* a witness — and for any balance exactly one party holds it. A sender cannot prove anything about the recipient's balance: it has no opening for it, and binding guarantees it cannot find another.

Thesis. Under commitment-based confidentiality, quantitative compliance cannot be transaction-gated. It must be **holder-attested** — the party that can prove is the party whose balance it is.

Not a weaker guarantee: a differently-shaped one. For thresholds that *qualify* a holder — the qualified-investor minimum, a concentration ceiling — it is exactly what the regulation asks.

What we built, on an unmodified upstream token

COMPONENT	WHAT IT IS
<code>velum-policy</code>	Adapter: Confidential Token's policy gate → RWA <code>IdentityVerifier</code> . One KYC registry governs the public asset and the confidential wrapper.
<code>disclose_balance_ge</code>	The predicate circuit upstream specifies (\$9) and does not ship — no public implementation exists in any repository, branch, tag or history.
<code>velum-attest</code>	On-chain verifier — a component the same spec marks out of scope . Every public input is read from the token and the regulatory profile; the prover supplies only the proof.
<code>circuits/seize</code>	Answers upstream's open question on individual clawback: premise verified, circuit built, and a partial seizure verified on-chain.

Measured, not estimated

Proof of position	0.25 s
Proof size · public inputs	14 592 B · 6
Seize proof · circuit cost	0.33 s · 93 opcodes
Quantitative rule inside transfer	+3 %
Circuit tests (4 packages)	69 passing
False claim	not witnessable
Full confidential flow, testnet	1 m 04 s

The problem, in their words

`COMPLIANCE.md:218` — on why individual clawback is unsolved:

"Because the clawback circuit does not have access to `vk_A` ... The follow-up revision will pin down whether `r_s` is supplied as a private witness with an auxiliary opening proof or **derived in-circuit from a separately escrowed value**."

They name the obstacle and list the remedy. We take that second option and say what to escrow: `vk_A` itself. Verified in code and on-chain: premise, circuit, and a seizure proof accepted by a Soroban verifier.

Every claim we make about their product is sourced to a file and line in Appendix A of the whitepaper — trait signatures and verbatim quotes, not our characterisation.

Anchor case

A Brazilian receivables fund (FIDC) under **CVM 175**, tokenized in senior and subordinated classes — **Plina Finance**. Today its register is public: any competitor reads who subscribed, how much, and when they left. The kit serves any regulated issuer with the same paradox.

STATUS · TESTNET ONLY

Built on

OpenZeppelin `stellar-contracts` (Confidential Token + RWA/T-REX), Noir/UltraHonk circuits, Soroban. Nothing upstream is modified.

Honest limits

Developer-preview primitives; the UltraHonk verifier is unaudited. Aggregates across accounts — concentration by tranche, subordination ratio — remain open.

Open source

MIT · `github.com/ThaisFReis/velum`
Whitepaper, circuits, contracts and reproduction steps in the repo.